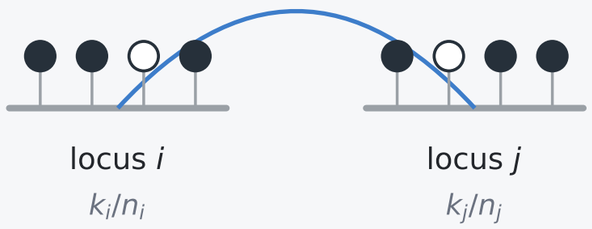
Long-range epigenetic concordance

a per-cell, per-bin signal that separates cell states even when mean methylation / accessibility is identical

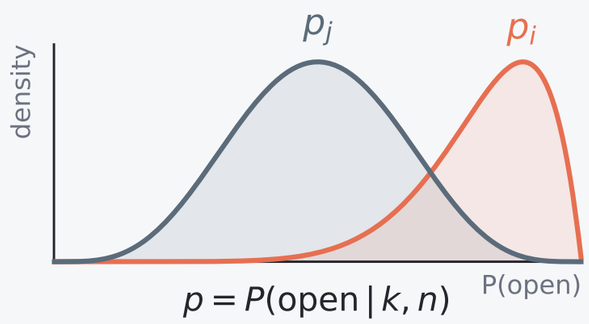
Single-cell NOMe-HiC contact

chromatin contact



GCH accessibility · HCG (CpG) methylation

Per-end state posterior



Beta–Binomial mixture (EM)

Soft pairwise co-state

co-open
$p_i\, p_j$

co-closed
$(1-p_i)(1-p_j)$

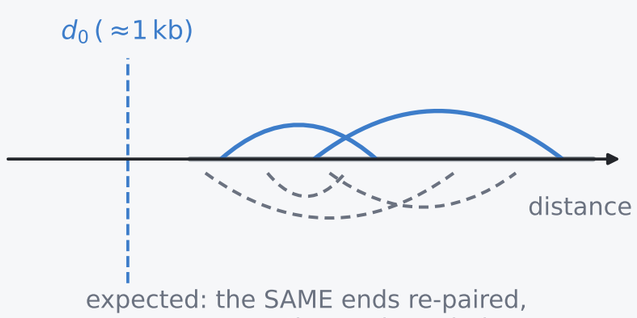
mixed
$p_i(1-p_j) + (1-p_i)p_j$

soft expected counts (no thresholding)

Distance-pooled expected

aggregate within 250-kb bin

observed: linked contacts



expected: the SAME ends re-paired,
every separation $\geq d_0$ pooled

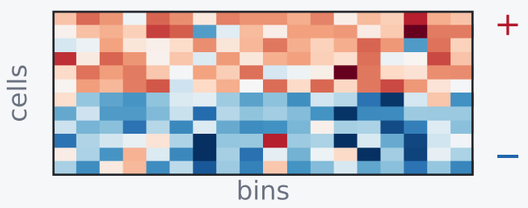
Long-range concordance

$$C = (O_{\text{open}} + O_{\text{closed}}) - (E_{\text{open}} + E_{\text{closed}})$$

$$F = \frac{O_{\text{open}}}{O_{\text{open}} + O_{\text{closed}}} \qquad F_{\text{exp}} = \frac{E_{\text{open}}}{E_{\text{open}} + E_{\text{closed}}}$$

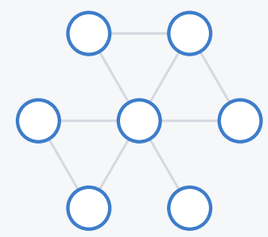
$$L = \operatorname{logit} F - \operatorname{logit} F_{\text{exp}} = \log \frac{O_{\text{open}}}{O_{\text{closed}}} - \log \frac{E_{\text{open}}}{E_{\text{closed}}}$$

C = magnitude · L = direction (marginal-free)



Cell-graph denoising (RWR)

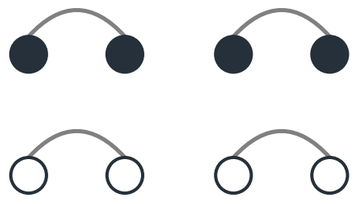
random-walk-with-restart smoothing
over the cell k-NN graph



quantile-normalised, imputed → UMAP

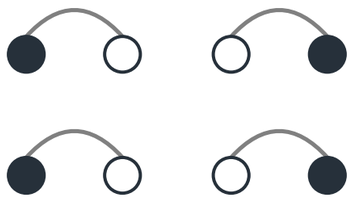
Same mean methylation / accessibility, opposite long-range concordance

cell state 1



concordance = 1.0

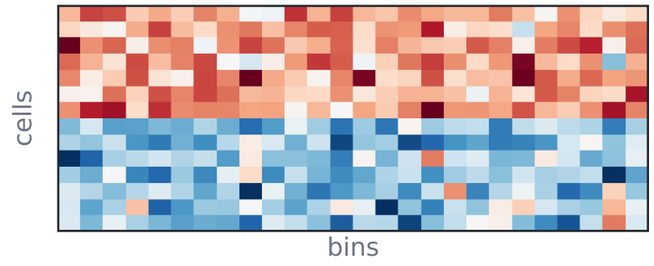
cell state 2



concordance = 0.0

workflow

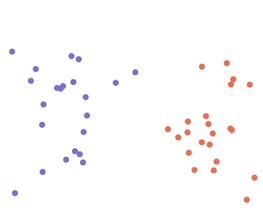
per-cell, per-bin C and L



the input to every analysis below

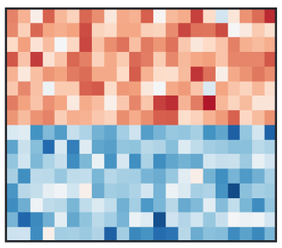
Downstream analyses

UMAP



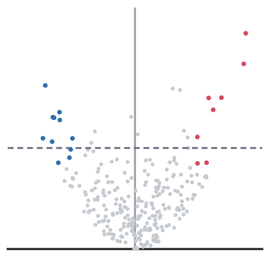
cell states separate

Heatmap



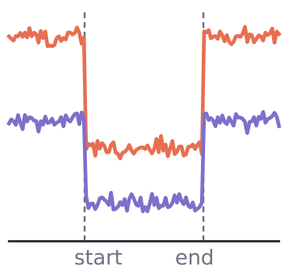
differential bins x cells

Volcano



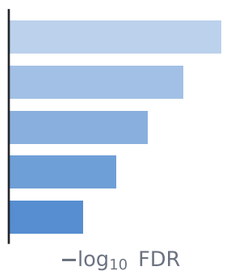
effect size vs FDR

Scaled profile



PMD / LAD / RT domains

KEGG / GO



genes in differential bins

● methylated / accessible ○ unmethylated / closed

each loop = two loci in contact within one cell